Instrument and Measurement Project Report

Electronic Distance Measuring Device(EDM)

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Overview

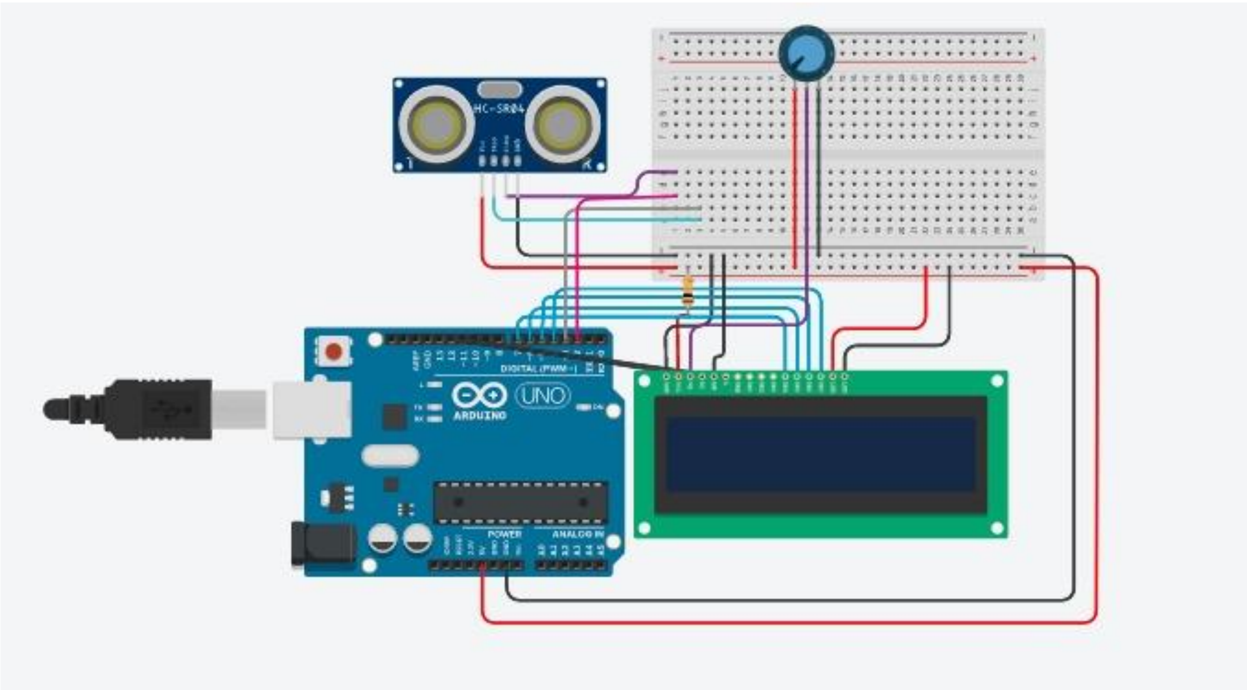
The group decided to make an EDM for this project. An EDM is the abbreviation of Electronic Distance Measurer; a device that can be used to place objects or points in three dimensions in relation to the unit. It can also be defined as a device used to determine the length between two points using electromagnetic waves. We are using several key components, one of which is an ultrasonic sensor; the sensor will measure the distance 20m to the target object by sending a sound pulse towards the target and measuring the time it takes for its echo to return.

The instrument is mostly used in the civil and construction sector. Another example of it is theodolite.

Component Used

- HC SR04--Ultrasonic sensor.
- LCD display 16x2.
- 9 Volts Battery.
- Arduino UNO board.
- Potentiometer
- Switch
- Vero board
- Battery
- Jumper Wires
- Resistors

Circuit Diagram



Methodology of the Circuit

The diagram shows a clear representation of how the circuit will look like and its working connections. We can see the Arduino UNO attached with the circuit as it contains the code for the whole project. We have an LCD display to exhibit the data gathered from the circuit and the sensor to measure distance. All units are connected with the help of jumper wires.

The following code was used for the Arduino:

```
#include <LiquidCrystal.h>

const int trigPin = 3;

const int echoPin = 2;

long duration;

double distance;

int op;
```

```
int rs = 12, e = 11, d4 = 7, d5 = 6, d6 = 5, d7 = 4;
LiquidCrystal lcd(rs, e, d4, d5, d6, d7);
void setup() {
  lcd.begin(16, 2);
  pinMode(trigPin, OUTPUT); // Sets the trigPin as an Output
  pinMode(echoPin, INPUT); // Sets the echoPin as an Input
  Serial.begin(9600);
  }
void loop() {
  lcd.setCursor(0, 0);

  // Clears the trigPin
digitalWrite(trigPin, LOW);
delayMicroseconds(2);
// Sets the trigPin on HIGH state for 10 micro seconds
digitalWrite(trigPin, HIGH);
//delayMicroseconds(10);
digitalWrite(trigPin, LOW);
// Reads the echoPin, returns the sound wave travel time in microseconds
duration = pulseIn(echoPin, HIGH);
// Calculating the distance,
distance= duration*0.0343/2;
// Prints the distance on the Serial Monitor
double cm=distance;
double m= cm/100;
double in= cm/2.54;
double ft= in/12;

Serial.print( cm );
Serial.print( "cm," );
Serial.print( in );
Serial.print( "in," );
Serial.print( ft );
```

```
Serial.print( "ft," );
Serial.print( m );
Serial.println( "m" );
delay(2000);
lcd.setCursor(0,0);
lcd.print( cm );
lcd.print( "cm," );
lcd.print( in );
lcd.print( "in," );
lcd.setCursor(0,1);
lcd.print( ft );
lcd.print( "ft," );
lcd.print( m );
lcd.print( "m" );
//delay(2000);
 }
```

The pin connections with the Arduino are given below:

Lcd:

Pin1 to ground

Pin2 to vcc

Pin3 V0 to potentiometer

Pin4 RS to 12

Pin5 Rw to gnd

Pin6 E to 11

Pin7-10 not use

Pin11 D4 to 7

Pin 12 D5 to 6

Pin13 D6 to 5

Pin14 D7 to 4

Pin15 A to vcc with 220 resistor

Pin16 k to gnd

HC-SRO4

Vcc to vcc

Gnd to gnd

Trig to 3

Echo to 4

The Circuit in Real-time